AADG classes: approximate string matching and string indexes

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1 Approximate string matching.

1. Inspect the code of the DP algorithm for approximate string matching.

2. How to modify the code to obtain an algorithm that finds all occurrences of the pattern with $\leq k$ edit operations?

2 Index assisted approximate string matching.

1. Inspect the code of the pattern partitioning algorithm for index-assisted approximate string matching.

2. We will say that the pattern partitioning is *efficient* iff each fragment `part` has the expected number of occurrences in the text < 1. Given genome sequence of length m, how long must be fragments `part` to meet this criterion?

3. Suppose you need to design an algorithm to map long reads from a new sequencing technology to the human genome. What is the largest read error rate for which the pattern partitioning approach is efficient?

4. How can you combine pattern partitioning with neighborhood search? Provide examples of parameters for which the combined approach is superior to the others.

3 Experimental verification.

1. Implement one of the algorithms finding all pattern occurrences with $\leq k$ edit operations:
   - modified DP algorithm,
   - FM-index assisted neighborhood search algorithm,
   - FM-index assisted pattern partitioning algorithm,
   - FM-index assisted pattern partitioning algorithm combined with edit distance 1-neighborhood generation.
2. Examine the efficiency of your implementation on the dataset from moodle.